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(54) **UNDERCUT ANCHOR**

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F16B 11/00 (2006.01)

F16B 13/08 (2006.01)

F16B 13/12 (2006.01)

(52) **U.S. Cl.**

CPC **E04B 1/4157** (2013.01); **E04B 1/4121** (2013.01); **F16B 11/008** (2013.01); **F16B 13/06** (2013.01); **F16B 13/066** (2013.01); **F16B 13/0833** (2013.01); **F16B 13/0858** (2013.01); **F16B 13/124** (2013.01); **F16B 13/126** (2013.01); **F16B 13/128** (2013.01)

(58) **Field of Classification Search**

CPC .. F16B 13/066; F16B 13/065; F16B 13/0833; F16B 13/0858; F16B 13/06; F16B 2013/007

USPC 411/55, 54, 54.1
See application file for complete search history.

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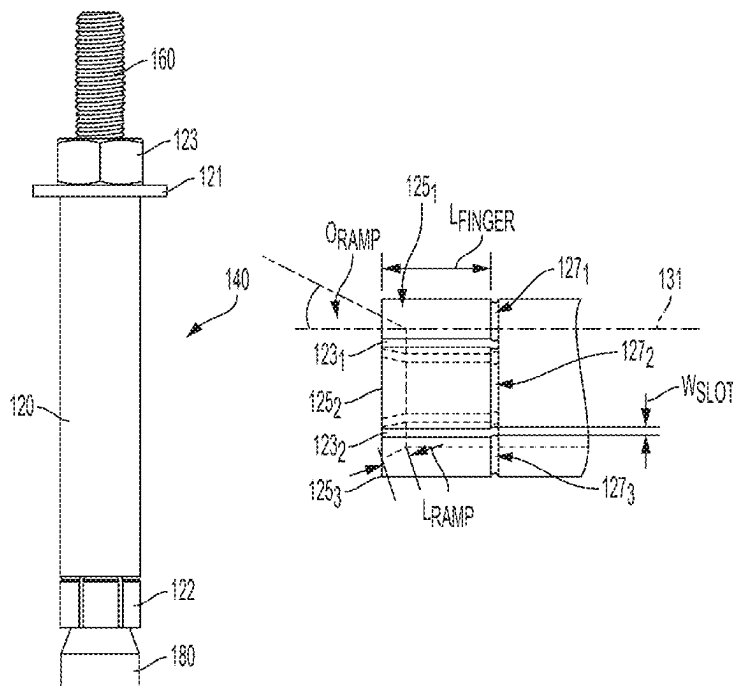
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(57) **ABSTRACT**

An undercut anchor having an at least partially threaded rod portion having an outer thread diameter D_{ROD} . The anchor having an expansion feature coupled to the rod portion and an expansion sleeve having an outer diameter $D_{OUTER\ SLEEVE}$, an inner diameter $D_{INNER\ SLEEVE}$ and a wall thickness T_{SLEEVE} . Furthermore, the expansion sleeve has a plurality of circumferentially distributed fingers which are configured to flex outwards when a ramp portion of length L_{RAMP} of each of the respective fingers is forced against the expansion feature during installation of the undercut anchor. The fingers each having a length L_{FINGER} and a distance therebetween of W_{SLOT} .

4 Claims, 7 Drawing Sheets



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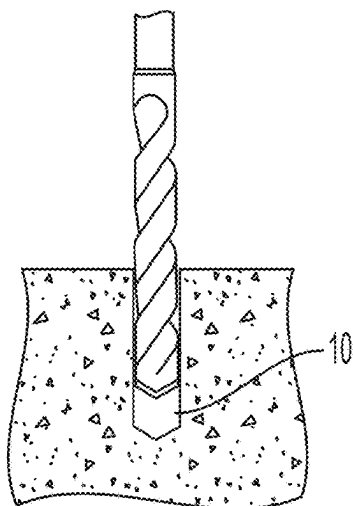


FIG. 1
(Prior Art)

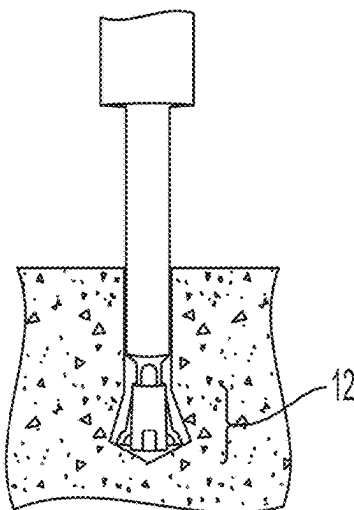


FIG. 2
(Prior Art)

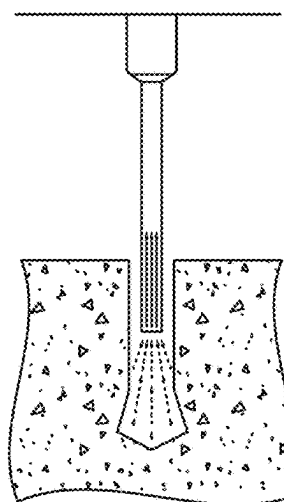


FIG. 3
(Prior Art)

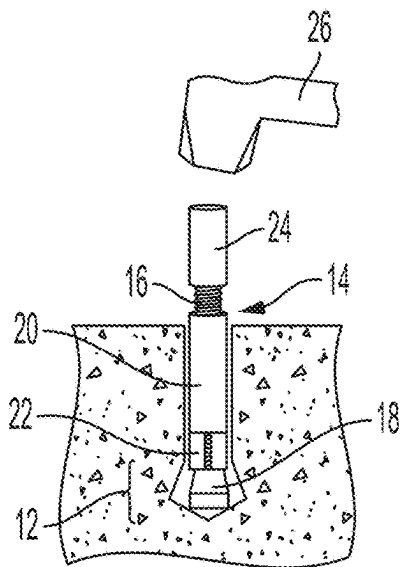


FIG. 4
(Prior Art)

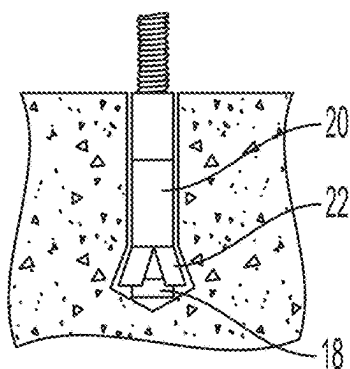


FIG. 5
(Prior Art)

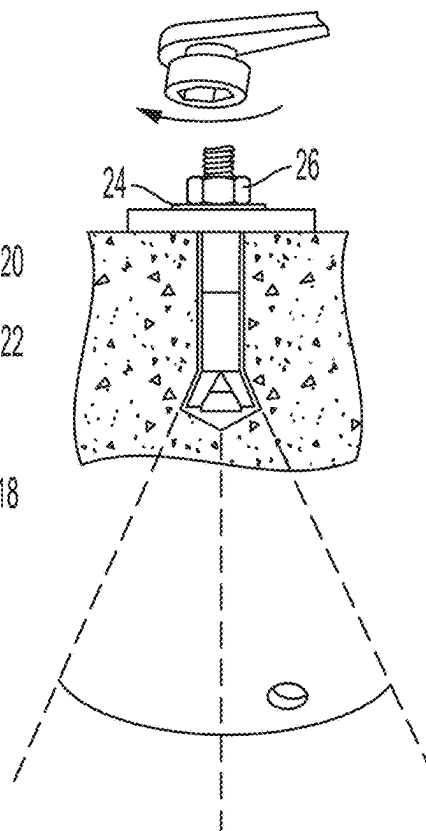


FIG. 6
(Prior Art)

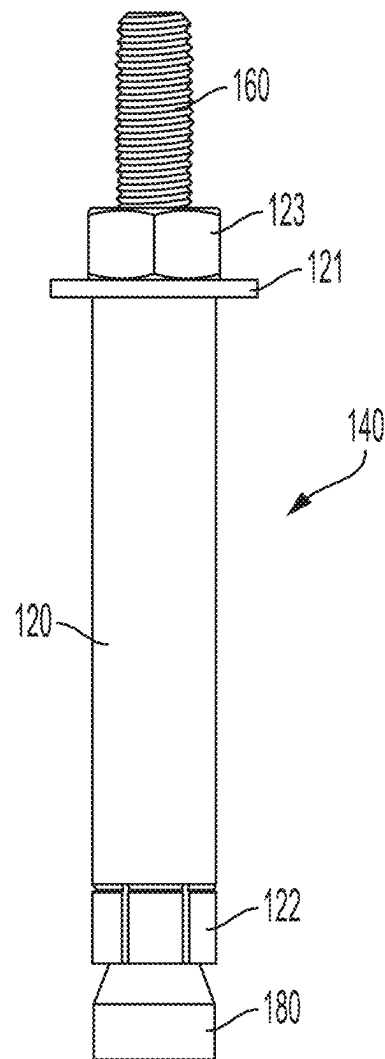


FIG. 7

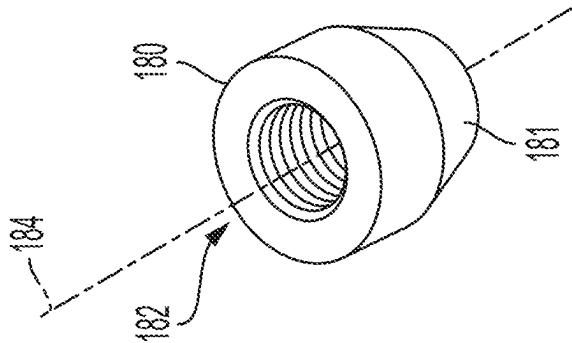


FIG. 8

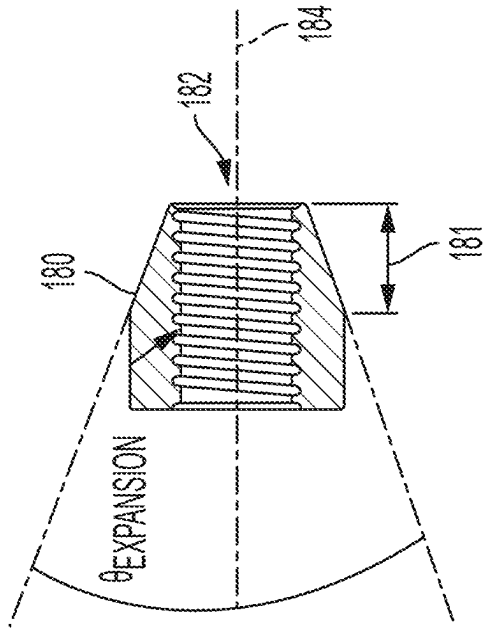


FIG. 9

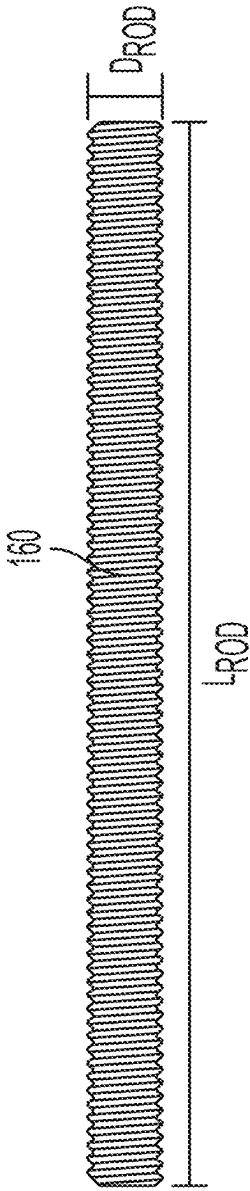


FIG. 10

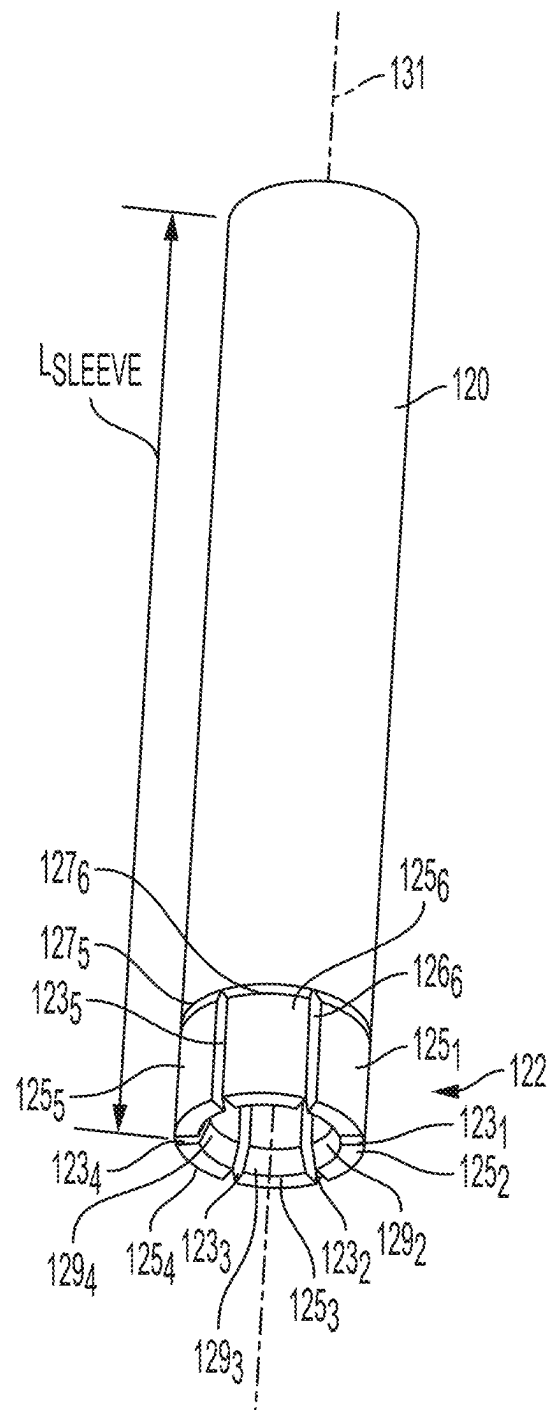


FIG. 11

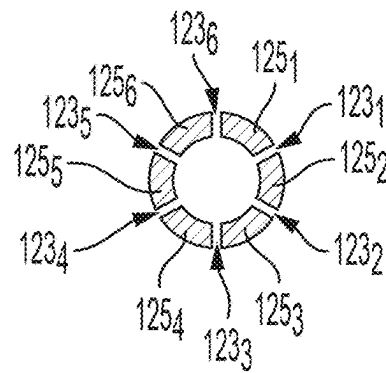


FIG. 12

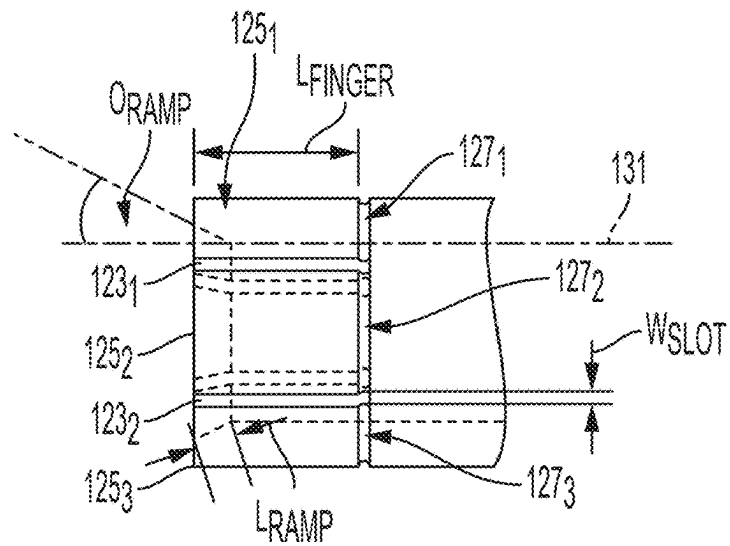


FIG. 13

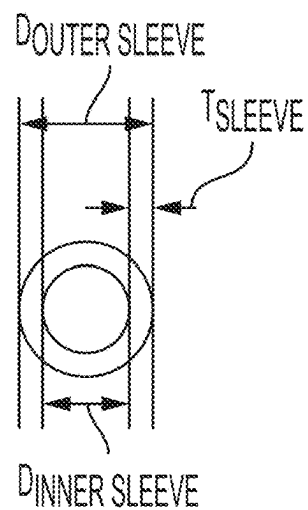


FIG. 14

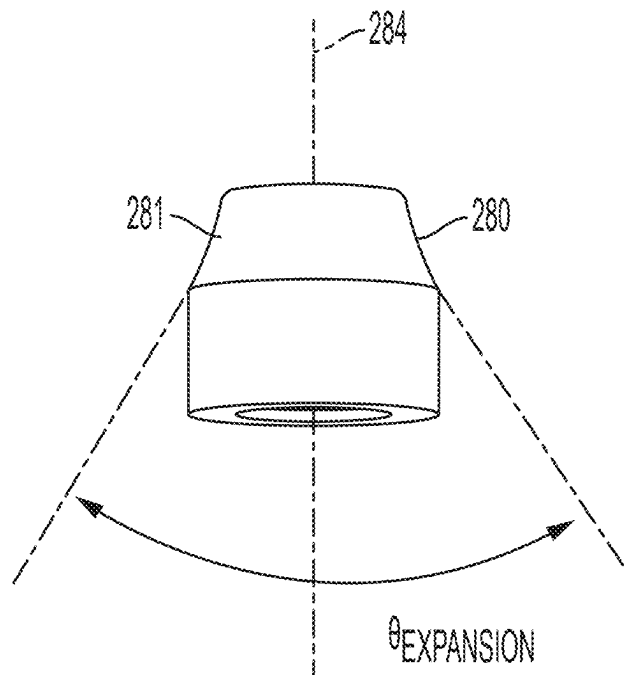


FIG. 15

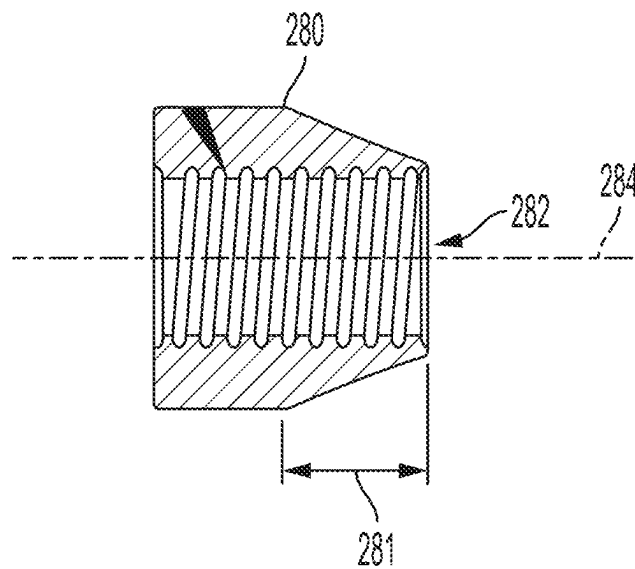


FIG. 16

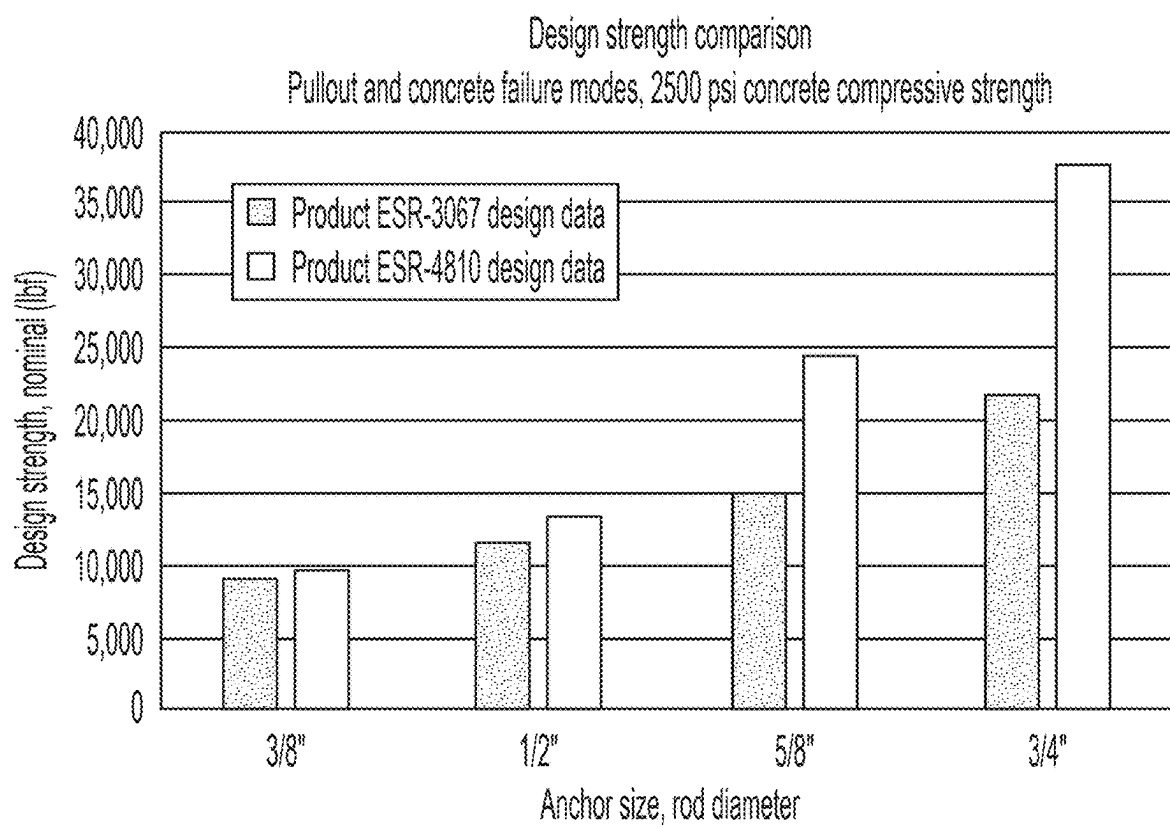


FIG. 17

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UNDERCUT ANCHOR

FIELD OF THE INVENTION

This specification concerns undercut anchors for securing objects to concrete.

BACKGROUND OF THE INVENTION

Undercut anchors are used for securing objects to a concrete surface, wherein undercut anchors have a higher pull out strength than other types of mechanical anchors such as drop-in anchors, screw anchors and expansion anchors.

FIGS. 1 to 6 show the steps involved in an undercut anchor installation method.

A hole 10 is drilled in a concrete surface using a hammer drill like in FIG. 1, wherein the hole 10 is generally cylindrical. Next an undercut region 12 is formed near the base of the hole 10 using an undercut drilling bit like in FIG. 2, wherein the undercut region 12 is essentially a conical section of the hole 10 which increases in diameter along an axis extending into the hole 10. The hole 10 is then cleaned by vacuuming out dust like in FIG. 3 and subsequently an undercut anchor 14 is inserted into the hole 10 like in FIG. 4. The undercut anchor 14 has a threaded rod section 16 fixed relative to an expansion feature 18. A sleeve 20 is located around the threaded rod section 16. The sleeve 20 at its end adjacent the expansion feature 18 has an expansion portion 22 defined by a series of axially extending, evenly circumferentially arranged, slots in the sleeve 20 for defining a plurality of fingers which can be caused to bend outwards by the expansion feature 18.

With further reference to FIG. 4 the external diameter of the sleeve 20 is selected to be slightly less than the internal diameter of the hole 10 such that upon inserting the anchor 14 into the hole 10 no friction is experienced between the inner surface of the hole 10 and the external surface of the sleeve 20. The anchor 14 is pushed into the hole 10 so that the expansion feature 18 touches the bottom of the hole 10 and so that the expansion portion 22 of the sleeve 20 extends into the undercut region 12. To facilitate movement of the expansion portion 22 into the undercut region 12 a cylindrical driver 24 is placed around the upper section of the threaded rod 16 and into engagement with the upper end of the sleeve 20. Impacting the cylindrical driver 24 with a hammer 26 for example pushes the sleeve 20 down into the hole, wherein such movement causes the expansion portion 22 to outwardly deform by cooperating with the expansion feature 18. The resulting arrangement is shown in FIG. 5. The length of the sleeve 20 is selected so that the sleeve 20 is located below the concrete surface when the undercut anchor 14 is installed. In through-bolt applications however the sleeve 20 will be long enough to extend from the hole after installation.

An object can then be fastened to the concrete using a nut 26 and washer 24 in conjunction with the installed undercut anchor. The upper part of the threaded rod 16 is caused to extend through a hole in the object to be secured. The washer 25, and subsequently the nut 26, are placed on the upper part of the threaded rod 16. Upon tightening the nut 26 against the washer 25, and thereby the object to be secured, the expansion feature 18 is urged in a pull-out direction, which further urges the expansion portion 22 against the conical surface defined by the undercut region 12. It will thus be appreciated that the expansion feature 18, the expansion

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portion 22 and the conical surface defined by the undercut region 12 of the hole 10 cooperate to resist pull-out of the undercut anchor 14.

The undercut angle θ illustrated in FIG. 6 is the angle between opposing surface sections of the conical surface defined by the undercut region 12. In other words, the undercut angle θ is basically the cone angle of the conical surface defined by the undercut region 12. Pull out strength of an undercut anchor is increased by increasing the undercut angle θ , however, until recently it has only been possible to form an undercut region 12 having an undercut angle θ in the range of 20 to 30 degrees due to limitations with drilling technology.

Some undercut anchors like the HDA-PTM and HDA-TTM (undercut anchors available from Hilti) form their own undercut region during installation. A drill is only required to form a cylindrical hole in a concrete surface for receiving the HDA-PTM and HDA-TTM undercut anchors. During installation, while drawing its expansion feature against its expansion sleeve, the HDA-PTM and HDA-TTM undercut anchors are themselves rotated whereby such rotation causes the anchor to cut its own undercut region in the concrete. Undercut angles θ capable of being achieved with the HDA-PTM and HDA-TTM undercut anchors can reach around 55 degrees, however, such anchors are highly engineered and so are complicated to manufacture compared to traditional undercut anchors which require a pre-drilled undercut region. What is needed therefore is a compromise, namely an undercut anchor solution that can achieve undercut angles θ greater than 30 degrees while remaining simpler to manufacture than an undercut anchor which forms its own undercut section during an installation process.

Due to improvements in drilling technology undercut drilling bits are now commercially available that can form undercut regions having undercut angles up to about 40 degrees (e.g., measured between opposite surfaces of the hold). As an example, the DEWALT® branded undercut drilling bits DFX21381, DFX21121, DFX21581 and DFX21341 can reliably drill undercut regions in concrete having an undercut angle up to about 40 degrees. Applicant has discovered that certain cleverly designed anchors may have significantly improved capacity when under cut holes have angles of 31 degrees or larger when measured between opposite sides of the cone shaped portion of the hole. For example, there are significant unexpected load capacity improvements between about 37 and about 43 degrees and between about 36 and about 43 degrees and between about 35 and about 43 and between about 34 and about 43 degrees.

Aspects of the present invention relate to an undercut anchor which is designed to function in holes having an undercut angle θ up to about 40 degrees (in particular, about 37 degrees to about 43 degrees).

BRIEF SUMMARY OF THE INVENTION

According to the present invention there is provided an undercut anchor including an at least partially threaded rod portion having an outer thread diameter D_{ROD} ; an expansion feature coupled to the rod portion; an expansion sleeve having an outer diameter $D_{OUTER\ SLEEVE}$, an inner diameter $D_{INNER\ SLEEVE}$ and a wall thickness T_{SLEEVE} . The expansion sleeve has a plurality of circumferentially distributed fingers which are configured to flex outwards when a ramp portion of length L_{RAMP} of each of the respective fingers is forced against the expansion feature during installation of the undercut anchor. The fingers each have a length L_{FINGER} and a distance therebetween of W_{SLOT} . The undercut anchor is

configured according to at least one of the following: the ratio of $D_{OUTER\ SLEEVE}$ to D_{ROD} is between 1.50 to 2.00; the ratio of $D_{INNER\ SLEEVE}$ to D_{ROD} is between 1.00 to 1.15; the ratio of T_{SLEEVE} to D_{ROD} is between 0.25 to 0.50; the ratio of L_{FINGER} to D_{ROD} is between 1.00 to 1.50; the ratio of L_{RAMP} to D_{ROD} is between 0.15 to 0.30; or the ratio of W_{SLOT} to D_{ROD} is between 0.05 to 0.20.

BRIEF DESCRIPTION OF THE DRAWINGS

Various aspects and embodiments of the invention will now be described by way of non-limiting example with reference to the accompanying drawings, in which:

FIGS. 1 to 6 are schematic illustrations of the steps involved in a prior art installation method for installing an undercut anchor.

FIG. 7 is a schematic illustration of an undercut anchor.

FIG. 8 is a schematic illustration of the expansion feature of the anchor in FIG. 7.

FIG. 9 is a schematic illustration of a cross-section of the expansion feature in FIG. 8.

FIG. 10 is a schematic illustration of the threaded rod section of the anchor in FIG. 7.

FIG. 11 is a schematic illustration of the sleeve of the anchor in FIG. 7.

FIG. 12 is a schematic illustration of the bottom end of the sleeve in FIG. 11.

FIG. 13 is a partially transparent schematic illustration of the bottom end of the sleeve in FIG. 11.

FIG. 14 is a schematic illustration of the top end of the sleeve in FIG. 11.

FIG. 15 is a schematic illustration of an alternative expansion feature.

FIG. 16 is a schematic cross-sectional illustration of the expansion feature in FIG. 15.

FIG. 17 is a table showing performance of the anchor in FIG. 7 relative to that of a prior art anchor.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

FIG. 7 illustrates an undercut anchor **140** which is designed to function in holes having an undercut angle θ up to about 40 degrees, wherein a washer **121** and nut **123** are shown attached to the undercut anchor **140**. The undercut anchor **140** functions in a similar way to the undercut anchor described in the background section and cooperates with the washer **121** and nut **123** to clamp an object to a concrete surface, however, certain dimensions of the undercut anchor **140** have been specifically chosen in order to achieve enhanced performance of the anchor **140** in holes having an undercut angle θ up to about 40 degrees. Therefore, an installation of the undercut anchor of the present invention may include a hole with an undercut angle θ of about 37 to about 43 degrees and/or may include an expansion feature with an expansion angle θ of about 37 to about 43 degrees. The undercut anchor **140** is made of steel.

The undercut anchor **140** can be made of carbon steel such as A36 carbon steel or B7 carbon steel. In some embodiments the carbon steel is zinc plated.

Alternatively, the undercut anchor **140** can be made of 316 stainless steel, wherein the 316 stainless steel version is particularly suitable for exterior use and other applications where a high level of corrosion resistance is required.

A threaded rod **160** is fixed relative to an expansion feature **180**. With reference to FIGS. 8 to 10 the threaded rod **160** has an outer thread diameter D_{ROD} and a length L_{ROD} .

The expansion feature **180** defines a conical portion **181** having a cone angle $\theta_{EXPANSION}$. The expansion feature **180** defines an opening **182** which extends along the major axis **184** of the expansion feature **180**. This opening **182** is threaded and thereby is capable of threadably connecting with the threaded rod section **160**. In order to secure the expansion feature **180** to the threaded rod section **160** adhesive is applied to the threaded internal surface of the expansion feature **180** before connecting it with the threaded rod section **160**.

Referring to FIG. 7 a cylindrical sleeve **120** is located around the threaded rod **160**. The end of the sleeve **120** adjacent the expansion feature **180** has an expansion portion **122**. FIGS. 11 to 14 show the sleeve **120** has an outer diameter $D_{OUTER\ SLEEVE}$, an inner diameter $D_{INNER\ SLEEVE}$, a wall thickness T_{SLEEVE} and a length L_{SLEEVE} .

The expansion portion **122** has a plurality of axially extending slots **123₁** to **123₆** evenly circumferentially arranged around the sleeve **120**, which defines a plurality of fingers **125₁** to **125₆**. The width of each slot **123₁** to **123₆**, in other words the distance between adjacent fingers **125₁** to **125₆** is W_{SLOT} . The fingers **125₁** to **125₆** have the same length L_{FINGER} . A weakened portion **127₁** to **127₆** of the sleeve **120** is located at the proximal end of each of the fingers **125₁** to **125₆**, only three of which are shown in FIG. 10. The weakened portions **127₁** to **127₆** enable the fingers **125₁** to **125₆** to flex outwards under force from the expansion feature **180** in use.

With particular reference to FIGS. 11 and 13 the inner distal end of each of the fingers **125₁** to **125₆** defines a ramped surface portion **129₁** to **129₆** only three of which are shown in FIG. 11. The length of each ramped surface portion **129₁** to **129₆** is L_{RAMP} . The angle θ_{RAMP} of the ramped surface portions **129₁** to **129₆** relative to the major axis **131** of the sleeve **120** is selected to match the slope of the conical portion **181** of the expansion feature **180**, wherein the angle θ_{RAMP} of the ramped surface portions **129₁** to **129₆** is therefore half the cone angle $\theta_{EXPANSION}$ of the conical portion **181**.

It is known to categorise different sized undercut anchors as $\frac{3}{8}$ ", $\frac{1}{2}$ ", $\frac{5}{8}$ " and $\frac{3}{4}$ " varieties, wherein these figures are indicative of the outer thread diameter D_{ROD} (these values being denoted in inches, namely 0.375", 0.5", 0.625" and 0.75"). For each of these four categories the applicant has determined from experimentation the optimum parameters listed in the table 1 below which give optimum use performance of the heretofore described undercut anchor **140** in holes in concrete with an undercut angle up to 40 degrees. Such enhanced performance is characterised by maximising pull-out resistance of the undercut anchor **140** for the respective dimension of the outer thread diameter D_{ROD} .

TABLE 1

D_{ROD} (Inches)	0.375	0.5	0.625	0.75
$D_{OUTER\ SLEEVE}$ (Inches)	0.6875	0.8125	1	1.25
$D_{INNER\ SLEEVE}$ (Inches)	0.405	0.535	0.665	0.79
T_{SLEEVE} (Inches)	0.14125	0.13875	0.1675	0.23
L_{FINGER} (Inches)	0.495	0.532	0.661	0.881
L_{RAMP} (Inches)	0.075	0.112	0.149	0.187
W_{SLOT} (Inches)	0.055	0.055	0.055	0.055

Parameters of the undercut anchor **140** not listed in the above table 1 can be varied depending on the usage context and specific user requirements. For example the length L_{SLEEVE} of the sleeve **120** (see FIG. 11) can be selected so that the sleeve **120** is totally located below the concrete surface in which the undercut anchor **140** is installed,

although in through-bolt applications the sleeve **120** may be required to protrude out of the hole in which the anchor **140** is received in order to extend through an opening in the object that is being fixed to a concrete surface. Also for each set of parameters listed in table 1 the cone angle $\theta_{EXPANSION}$ of the conical portion **181** can be adjusted, wherein it may range between 24 and 56 degrees and may for example be 40 degrees. The reader is reminded that the angle θ_{RAMP} of the ramped surface portions **129₁** to **129₆** relative to the major axis **131** of the sleeve **120** is selected to match the slope of the conical portion **181** of the expansion feature **180**, wherein the angle θ_{RAMP} of the ramped surface portions **129₁** to **129₆** is therefore half the cone angle $\theta_{EXPANSION}$ of the conical portion **181**.

Table 2 below expresses the experimentally determined ranges of parameters which give acceptable use performance of the undercut anchor **140** in holes in concrete with an undercut angle up to about 40 degrees. Such acceptable use performance is characterised by maximising pull-out resistance of the undercut anchor **140** for the selected dimension of the outer thread diameter D_{ROD} .

TABLE 2

	Minimum	Maximum
$D_{OUTER\ SLEEVE}/D_{ROD}$	1.50	2.00
$D_{INNER\ SLEEVE}/D_{ROD}$	1.00	1.15
T_{SLEEVE}/D_{ROD}	0.25	0.50
L_{FINGER}/D_{ROD}	1.00	1.50
L_{RAMP}/D_{ROD}	0.15	0.30
W_{SLOT}/D_{ROD}	0.05	0.20

It is here stated that the respective sets of unexpected, parameters in Table 1 for the different anchor sizes fall within the scope of the range expressed in Table 2.

Performance data of the heretofore described undercut anchor **140** relative to a prior art undercut anchor is illustrated in FIG. 17. This drawing shows for such anchors the design strength, nominal vs. different sized versions of such undercut anchors, namely $\frac{3}{8}$ ", $\frac{1}{2}$ ", $\frac{5}{8}$ " and $\frac{3}{4}$ " varieties wherein as before these figures are indicative of the outer thread diameter D_{ROD} . Some explanation of the data in FIG. 17 is as follows. ICC Evaluation Service (ICC-ES) issues technical approvals for anchors in order to facilitate their launch in the United States. The ICC-ES report for the undercut anchor **140** described in this specification which is optimised for use in holes having an undercut angle θ up to about 40 degrees is ICC-ES report number ESR-4810 issued in June 2021 such report being published by and available from ICC-ES here <https://icc-es.org/report-listing/esr-4810/> and wherein the contents of this report are incorporated herein by reference. The ICC-ES report for a prior art undercut anchor which is not optimised for use in holes having an undercut angle θ up to about 40 degrees is ICC-ES report number ESR-3067 such report being published by and available from ICC-ES here <https://icc-es.org/report-listing/esr-3067/> and wherein the contents of this report are incorporated herein by reference. The undercut anchor which is the subject of ICC-ES report number ESR-3067 is optimised for use in holes having an undercut angle θ up to about 22 degrees. Based on the information set out in ICC-ES report numbers ESR-3067 and ESR-4810 a person skilled in the art will be able to calculate the information illustrated in FIG. 17 without exercising inventive activity.

It will be appreciated that whilst various aspects and embodiments have heretofore been described, the scope of the present invention is not limited thereto and instead

extends to encompass all arrangements, and modifications and alterations thereto, which fall within the spirit and scope of the appended claims.

In some embodiments the expansion feature **180** has a different shape. FIGS. 15 and 16 show an alternative expansion feature **280** wherein corresponding reference numerals increased by 100 indicate like features in the embodiment of FIGS. 8 and 9. The difference between the previously described expansion feature **180** and the alternative expansion feature **280** is that the conical section **281** has an outer diameter which increases in a non-linear manner from one end of the conical section **281** to the other. At the narrowest part of the conical section **281** the cone angle defined by the outer surface is small, wherein the cone angle increases based on distance along the major axis **284** of the expansion feature **280**. At the widest part of the conical section **281** the cone angle $\theta_{EXPANSION}$ corresponds in dimension to that of the previously described expansion feature **180**. This trumpet-shaped expansion feature **280** provides that less force is required to be imparted between the sleeve **120** and expansion feature **280** in order for the expansion feature **280** to outwardly deform the fingers **125₁** to **125₆** of the expansion portion **122**.

In some embodiments the expansion feature **180**, **280** is welded to the threaded rod section **160** after such features are threadably engaged instead of using adhesive.

In some embodiments the expansion feature **180**, **280** does not have a threaded opening **182** and is welded to the threaded rod section **160**.

The threaded rod section **160** does not need to be threaded along its whole length and only needs to be threaded sufficiently far along its length in order to enable a nut to be threadably engaged with it in order to secure an object to a concrete surface after anchor installation.

In some embodiments expansion sleeve **120** may have more or less than six fingers **125** provided that the dimensions of L_{FINGER} , L_{RAMP} and W_{SLOT} remain within the ranges set out in table 2.

What is claimed is:

1. An undercut anchor comprising: an at least partially threaded rod portion having an outer thread diameter D_{ROD} ; an expansion feature coupled to the rod portion; an expansion sleeve having an outer diameter $D_{OUTER\ SLEEVE}$, an inner diameter $D_{INNER\ SLEEVE}$ and a wall thickness T_{SLEEVE} wherein the expansion sleeve has a plurality of circumferentially distributed fingers which are configured to flex outwards when a ramp portion of length L_{RAMP} of each of the respective fingers is forced against the expansion feature during installation of the undercut anchor, the fingers each having a length L_{FINGER} and a distance therebetween of W_{SLOT} ; wherein the undercut anchor is configured according to at least one of the following:

the ratio of $D_{OUTER\ SLEEVE}$ to D_{ROD} is between 1.50 to 2.00;

the ratio of $D_{INNER\ SLEEVE}$ to D_{ROD} is between 1.00 to 1.15;

the ratio of T_{SLEEVE} to D_{ROD} is between 0.25 to 0.50;

the ratio of L_{FINGER} to D_{ROD} is between 1.00 to 1.50;

the ratio of L_{RAMP} to D_{ROD} is between 0.15 to 0.30;

the ratio of W_{SLOT} to D_{ROD} is between 0.05 to 0.20, and wherein the plurality of circumferentially distributed fingers are flexed outward to form an expansion angle between about 37° to about 43°, the undercut anchor configured to the following dimensions:

D_{ROD} (Inches)	0.375
$D_{OUTER\ SLEEVE}$ (Inches)	0.6875
$D_{INNER\ SLEEVE}$ (Inches)	0.405
T_{SLEEVE} (Inches)	0.14125
L_{FINGER} (Inches)	0.495
L_{RAMP} (Inches)	0.075
W_{SLOT} (Inches)	0.055.

2. An undercut anchor comprising: an at least partially threaded rod portion having an outer thread diameter D_{ROD} ; an expansion feature coupled to the rod portion; an expansion sleeve having an outer diameter $D_{OUTER\ SLEEVE}$, an inner diameter $D_{INNER\ SLEEVE}$ and a wall thickness T_{SLEEVE} wherein the expansion sleeve has a plurality of circumferentially distributed fingers which are configured to flex outwards when a ramp portion of length L_{RAMP} of each of the respective fingers is forced against the expansion feature during installation of the undercut anchor, the fingers each having a length L_{FINGER} and a distance therebetween of W_{SLOT} ; wherein the undercut anchor is configured according to at least one of the following:

the ratio of $D_{OUTER\ SLEEVE}$ to D_{ROD} is between 1.50 to 2.00;

the ratio of $D_{INNER\ SLEEVE}$ to D_{ROD} is between 1.00 to 1.15;

the ratio of T_{SLEEVE} to D_{ROD} is between 0.25 to 0.50;

the ratio of L_{FINGER} to D_{ROD} is between 1.00 to 1.50;

the ratio of L_{RAMP} to D_{ROD} is between 0.15 to 0.30;

the ratio of W_{SLOT} to D_{ROD} is between 0.05 to 0.20, and wherein the plurality of circumferentially distributed fingers are flexed outward to form an expansion angle between about 37° to about 43°, the undercut anchor configured to the following dimensions:

D_{ROD} (Inches)	0.5
$D_{OUTER\ SLEEVE}$ (Inches)	0.8125
$D_{INNER\ SLEEVE}$ (Inches)	0.535
T_{SLEEVE} (Inches)	0.13875
L_{FINGER} (Inches)	0.532
L_{RAMP} (Inches)	0.112
W_{SLOT} (Inches)	0.055.

3. An undercut anchor comprising: an at least partially threaded rod portion having an outer thread diameter D_{ROD} ; an expansion feature coupled to the rod portion; an expansion sleeve having an outer diameter $D_{OUTER\ SLEEVE}$, an inner diameter $D_{INNER\ SLEEVE}$ and a wall thickness T_{SLEEVE} wherein the expansion sleeve has a plurality of circumferentially distributed fingers which are configured to flex outwards when a ramp portion of length L_{RAMP} of each of the respective fingers is forced against the expansion feature during installation of the undercut anchor, the fingers each having a length L_{FINGER} and a distance therebetween of W_{SLOT} ; wherein the undercut anchor is configured according to at least one of the following:

the ratio of $D_{OUTER\ SLEEVE}$ to D_{ROD} is between 1.50 to 2.00;

the ratio of $D_{INNER\ SLEEVE}$ to D_{ROD} is between 1.00 to 1.15;

the ratio of T_{SLEEVE} to D_{ROD} is between 0.25 to 0.50;

the ratio of L_{FINGER} to D_{ROD} is between 1.00 to 1.50;

the ratio of L_{RAMP} to D_{ROD} is between 0.15 to 0.30;

the ratio of W_{SLOT} to D_{ROD} is between 0.05 to 0.20, and wherein the plurality of circumferentially distributed fingers are flexed outward to form an expansion angle between about 37° to about 43°, the undercut anchor configured to the following dimensions:

D_{ROD} (Inches)	0.625
$D_{OUTER\ SLEEVE}$ (Inches)	1
$D_{INNER\ SLEEVE}$ (Inches)	0.665
T_{SLEEVE} (Inches)	0.1675
L_{FINGER} (Inches)	0.661
L_{RAMP} (Inches)	0.149
W_{SLOT} (Inches)	0.055.

4. An undercut anchor comprising: an at least partially threaded rod portion having an outer thread diameter D_{ROD} ; an expansion feature coupled to the rod portion; an expansion sleeve having an outer diameter $D_{OUTER\ SLEEVE}$, an inner diameter $D_{INNER\ SLEEVE}$ and a wall thickness T_{SLEEVE} wherein the expansion sleeve has a plurality of circumferentially distributed fingers which are configured to flex outwards when a ramp portion of length L_{RAMP} of each of the respective fingers is forced against the expansion feature during installation of the undercut anchor, the fingers each having a length L_{FINGER} and a distance therebetween of W_{SLOT} ; wherein the undercut anchor is configured according to at least one of the following:

the ratio of $D_{OUTER\ SLEEVE}$ to D_{ROD} is between 1.50 to 2.00;

the ratio of $D_{INNER\ SLEEVE}$ to D_{ROD} is between 1.00 to 1.15;

the ratio of T_{SLEEVE} to D_{ROD} is between 0.25 to 0.50;

the ratio of L_{FINGER} to D_{ROD} is between 1.00 to 1.50;

the ratio of L_{RAMP} to D_{ROD} is between 0.15 to 0.30;

the ratio of W_{SLOT} to D_{ROD} is between 0.05 to 0.20, and wherein the plurality of circumferentially distributed fingers are flexed outward to form an expansion angle between about 37° to about 43°, the undercut anchor configured to the following dimensions:

D_{ROD} (Inches)	0.75
$D_{OUTER\ SLEEVE}$ (Inches)	1.25
$D_{INNER\ SLEEVE}$ (Inches)	0.79
T_{SLEEVE} (Inches)	0.23
L_{FINGER} (Inches)	0.881
L_{RAMP} (Inches)	0.187
W_{SLOT} (Inches)	0.055.

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